

CHAPTER 6

Anatomy of Flowering Plants

Meristematic Tissues

- Consider the following plant tissues: (2023-Manipur)
 A. Axillary buds B. Fascicular vascular cambium
 C. Interfascicular cambium D. Cork cambium
 E. Intercalary meristem
 Identify the lateral meristems among the above.
 a. (A), (C) and (D) only
 b. (B), (C) and (D) only
 c. (A), (B), (C) and (E) only
 d. (A), (B), (D) and (E) only

Permanent Tissues

- Which of the following simple tissues are commonly found in the fruit walls of nuts and pulp of pear? (2024 Re)
 a. Sclereids b. Fibres
 c. Parenchyma d. Collenchyma
- Which one of the following is **not** found in Gymnosperms? (2024 Re)
 a. Sieve cells b. Albuminous cells
 c. Tracheids d. Vessels
- Which of the following helps in maintenance of the pressure gradient in sieve tubes? (2024 Re)
 a. Albuminous cells b. Sieve cells
 c. Phloem parenchyma d. Companion cells
- Given below are two statements: (2024)
Statement-I: Parenchyma is living but collenchyma is dead tissue.
Statement-II: Gymnosperms lack xylem vessels but presence of xylem vessels is the characteristic of angiosperms.
 In the light of the above statements, choose the correct answer from the options given below:
 a. Statement-I is true but Statement-II is false
 b. Statement-I is false but Statement-II is true
 c. Both Statement-I and Statement-II are true
 d. Both Statement-I and Statement-II are false
- Given below are two statements: (2023)
Statement-I: Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.
Statement-II: Exarch condition is the most common feature of the root system.

In the light of the above statements, choose the correct answer from the options given below:

- Statement-I is incorrect but Statement-II is true.
 b. Both Statement-I and Statement-II are true.
 c. Both Statement-I and Statement-II are false.
 d. Statement-I is correct but Statement-II is false.
- The type of tissue commonly found in the fruit wall of nuts is (2022 Re)
 a. Sclereid b. Parenchyma
 c. Collenchyma d. Sclerenchyma
- Read the following statements about the vascular bundles: (2022)
 A. In roots, xylem and phloem in a vascular bundle are arranged in an alternate manner along the different radii
 B. Conjoint closed vascular bundles do not possess cambium
 C. In open vascular bundles, cambium is present in between xylem and phloem
 D. The vascular bundles of dicotyledonous stem possess endarch protoxylem
 E. In monocotyledonous root, usually there are more than six xylem bundles present
 Choose the correct answer from the options given below.
 a. A, C, D and E only
 b. A, B and D only
 c. B, C, D and E only
 d. A, B, C and D only
- In old trees the greater part of secondary xylem is dark brown and resistant to insect attack due to (2022)
 A. Secretion of secondary metabolites and their deposition in the lumen of vessels
 B. Deposition of organic compounds like tannins and resins in the central layers of stem
 C. Deposition of suberin and aromatic substances in the outer layer of stem
 D. Deposition of tannins, gum, resin and aromatic substances in the peripheral layers of stem
 E. Presence of parenchyma cells, functionally active xylem elements and essential oils.
 Choose the correct answer from the options given below.
 a. B and D only b. A and B only
 c. C and D only d. D and E only

10. Match List-I with List-II

(2021)

	List-I		List-II
A.	Cells with active cell division capacity	(i)	Vascular tissues
B.	Tissue having all cells similar in structure and function	(ii)	Meristematic tissue
C.	Tissue having different types of cells	(iii)	Sclereids
D.	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple Tissue

Select the correct answer from the options given below.

- a. A-iv B-iii C-ii D-i
 b. A-i B-ii C-iii D-iv
 c. A-iii B-ii C-iv D-i
 d. A-ii B-iv C-i D-iii

11. The transverse section of a plant shows following anatomical features: (2020)

1. Large number of scattered vascular bundles surrounded by bundle sheath.
2. Large conspicuous parenchymatous ground tissue.
3. Vascular bundles conjoint and closed.
4. Phloem parenchyma absent.

Identify the category of plant and its part:

- a. Monocotyledonous root
 b. Dicotyledonous stem
 c. Dicotyledonous root
 d. Monocotyledonous stem

12. Phloem in gymnosperms lacks (2019)

- a. Albuminous cells and sieve cells
 b. Sieve tubes only
 c. Companion cells only
 d. Both sieve tubes and companion cells

13. Which of the following is made up of dead cells? (2017-Delhi)

- a. Xylem parenchyma b. Collenchyma
 c. Phellem d. Phloem

14. The balloon-shaped structures called tyloses: (2016 - II)

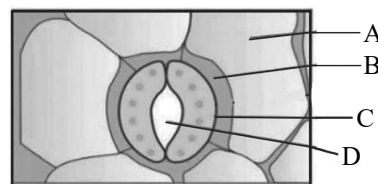
- a. Are extensions of xylem parenchyma cells into vessels
 b. Are linked to the ascent of sap through xylem vessels
 c. Originate in the lumen of vessels
 d. Characterise the sapwood

15. Tracheids differ from other tracheary elements in: (2014)

- a. Being lignified b. Having casparian strips
 c. Being imperforate d. Lacking nucleus

Tissue System

16. In the given figure, which component has thin outer walls and highly thickened inner walls? (2024)



- a. A b. B c. C d. D

17. Stomata in grass leaf are: (2018)

- a. Dumb-bell shaped b. Kidney shaped
 c. Rectangular d. Barrel shaped

18. Cortex is the region found between: (2016 - II)

- a. Endodermis and pith
 b. Endodermis and vascular bundle
 c. Epidermis and stele
 d. Pericycle and endodermis

19. Specialised epidermal cells surrounding the guard cells are called (2016 - I)

- a. Complementary cells b. Subsidiary cells
 c. Bulliform cells d. Lenticels

20. Vascular bundles in monocotyledons are considered closed because: (2015)

- a. There are no vessels with perforations
 b. Xylem is surrounded all around by phloem
 c. A bundle sheath surrounds each bundle
 d. Cambium is absent

Anatomy of Dicot & Monocot Plants

21. Given below are two statements: (2024 Re)

Statement-I: In a dicotyledonous leaf, the adaxial epidermis generally bears more stomata than the abaxial epidermis.

Statement-II: In a dicotyledonous leaf, the adaxially placed palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other.

In the light of the above statements, choose the correct answer from the options given below:

- a. Statement-I is true but Statement-II is false
 b. Statement-I is false but Statement-II is true
 c. Both Statement-I and Statement-II are true
 d. Both Statement-I and Statement-II are false

22. Bulliform cells are responsible for (2024)

- a. Increased photosynthesis in monocots.
 b. Providing large spaces for storage of sugars.
 c. Inward curling of leaves in monocots.
 d. Protecting the plant from salt stress.

23. The transverse section of a plant part showed polyarch, radial and exarch xylem, with endodermis and pericycle. The plant part is identified as: (2023-Manipur)

- a. Monocot root b. Dicot root
 c. Dicot stem d. Monocot stem

24. Initiation of lateral roots and vascular cambium during secondary growth takes place in cells of (2022 Re)

a. Pericycle b. Epiblema
c. Cortex d. Endodermis

25. Select the correct pair. (2021)

a.	In dicot leaves, vascular bundles are surrounded by large thick-walled cells	-	Conjunctive tissue
b.	Cells of medullary rays that form part of cambial rings	-	Interfascicular cambium
c.	Loose parenchyma cells rupturing the epidermis and forming a lens-shaped opening in bark	-	Spongy parenchyma
d.	Large colorless empty cells in the epidermis of grass leaves	-	Subsidiary cells

26. Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are (2020-Covid)

a. Guard cells b. Bundle sheath cells
c. Bulliform cells d. Lenticels

27. Grass leaves curl inwards during very dry weather. Select the most appropriate reason from the following (2019)

a. Closure of stomata
b. Flaccidity of bulliform cells
c. Shrinkage of air spaces in spongy mesophyll
d. Tyloses in vessels

28. Secondary xylem and phloem in dicot stem are produced by (2018)

a. Apical meristem b. Vascular cambium
c. Phellogen d. Axillary meristems

29. Casparian strips occur in (2018)

a. Epidermis b. Pericycle
c. Cortex d. Endodermis

30. A major characteristic of the monocot root is the presence of: (2015)

a. Vasculature without cambium
b. Cambium sandwiched between phloem and xylem along the radius
c. Open vascular bundles
d. Scattered vascular bundles

Secondary Growth

31. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R: (2023)

Assertion (A): Late wood has fewer xylary elements with narrow vessels.

Reason (R): Cambium is less active in winters.

In the light of the above statements, choose the correct answer from the options given below:

a. A is false but R is true.
b. Both A and R are true and R is the correct explanation of A.

- c. Both A and R are true but R is NOT the correct explanation of A.

d. A is true but R is false.

32. Identify the correct statements: (2023)

A. Lenticels are the lens-shaped openings permitting the exchange of gases.
B. Bark formed early in the season is called hard bark.
C. Bark is a technical term that refers to all tissues exterior to vascular cambium.
D. Bark refers to periderm and secondary phloem.
E. Phellogen is single-layered in thickness.

Choose the correct answer from the options given below:

a. B and C only b. B, C and E only
c. A and D only d. A, B and D only

33. Consider the following tissues in the stellar region of a stem showing secondary growth. (2023-Manipur)

(A) Primary xylem (B) Secondary xylem
(C) Primary phloem (D) Secondary phloem

Arrange these in the **correct** sequence of their position from pith towards cortex.

a. (A), (B), (D), (C) b. (B), (A), (C), (D)
c. (A), (B), (C), (D) d. (B), (A), (D), (C)

34. Interfascicular cambium is present between (2022 Re)

a. Secondary xylem and secondary phloem
b. Primary xylem and primary phloem
c. Pericycle and endodermis Two vascular bundles
d. Two vascular bundles

35. The anatomy of springwood shows some peculiar features. Identify the correct set of statements about springwood.

A. It is also called as the earlywood
B. In spring season cambium produces xylem elements with narrow vessels
C. It is lighter in colour.
D. The springwood along with autumnwood shows alternate concentric rings forming annual rings
E. It has lower density

Choose the correct answer from the options given below.

a. C, D and E only b. A, B, D and E only
c. A, C, D and E only d. A, B and D only

36. Match List-I with List-II (2021)

	List-I		List-II
A.	Lenticels	(i)	Phellogen
B.	Cork cambium	(ii)	Suberin deposition
C.	Secondary cortex	(iii)	Exchange of gases
D.	Cork	(iv)	Phelloderm

Choose the correct answer from the options given below:

a. A-iii B-i C-iv D-ii
b. A-ii B-iii C-iv D-i
c. A-iv B-ii C-i D-iii
d. A-iv B-i C-iii D-ii

37. Identify the **incorrect** statement. (2020)
- Sapwood is involved in conduction of water and minerals from root to leaf.
 - Sapwood is the innermost secondary xylem and is lighter in colour.
 - Due to deposition of tannins, resins, oils, etc., heart wood is dark in colour.
 - Heart wood does not conduct water but gives mechanical support.
38. Which of the following statements about cork cambium is **incorrect**? (2020-Covid)
- It forms a part of periderm
 - It is responsible for the formation of lenticels
 - It is a couple of layers thick
 - It forms secondary cortex on its outside
39. Which of the statements given below is **not** true about formation of annual rings in trees? (2019)
- Annual ring is a combination of spring wood and autumn wood produced in a year
 - Differential activity of cambium causes light and dark bands of tissue-early and late wood respectively.
 - Activity of cambium depends upon variation in climate.
 - Annual rings are not prominent in trees of temperate region.
40. Plants having little or no secondary growth are (2018)
- Grasses
 - Deciduous angiosperms
 - Conifers
 - Cycads
41. The vascular cambium normally gives rise to: (2017-Delhi)
- Phelloderm
 - Primary phloem
 - Secondary xylem
 - Periderm
42. Identify the wrong statement in context of heartwood: (2017-Delhi)
- Organic compounds are deposited in it
 - It is highly durable
 - It conducts water and minerals efficiently
 - It comprises dead elements with highly lignified wall
43. Read the different components from (A) to (D) in the list given below and tell the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem: (2015 Re)
- A. Secondary cortex B. Wood
C. Secondary phloem D. Phellem
- The correct order is:
- (A), (B), (D), (C)
 - (D), (A), (C), (B)
 - (D), (C), (A), (B)
 - (C), (D), (B), (A)
44. You are given a fairly old piece of dicot stem and a dicot root. Which of the following anatomical structures will you use to distinguish between the two? (2014)
- Cortical cells
 - Secondary xylem
 - Secondary phloem
 - Protoxylem
45. Lenticels are involved in: (2013)
- Photosynthesis
 - Transpiration
 - Gaseous exchange
 - Food transport
46. Interfascicular cambium develops from the cells of: (2013)
- Pericycle
 - Medullary rays
 - Xylem parenchyma
 - Endodermis
47. Age of a tree can be estimated by: (2013)
- Diameter of its heartwood
 - Its height and girth
 - Biomass
 - Number of annual rings

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|-----------|---------|---------|
| 1. (b) | 2. (a) | 3. (d) | 4. (d) | 5. (b) | 6. (a) | 7. (a) | 8. (None) | 9. (b) | 10. (d) |
| 11. (d) | 12. (d) | 13. (c) | 14. (a) | 15. (c) | 16. (c) | 17. (a) | 18. (c) | 19. (b) | 20. (d) |
| 21. (b) | 22. (c) | 23. (a) | 24. (a) | 25. (b) | 26. (c) | 27. (b) | 28. (b) | 29. (d) | 30. (a) |
| 31. (b) | 32. (c) | 33. (a) | 34. (d) | 35. (c) | 36. (a) | 37. (b) | 38. (d) | 39. (d) | 40. (a) |
| 41. (c) | 42. (c) | 43. (b) | 44. (d) | 45. (c) | 46. (b) | 47. (d) | | | |